

TraFoo: An Android Application for Food Delivery in Train

Md. Mosfikur Rahman
Department of Computer Science and
Engineering
Daffodil International University
Dhaka, Bangladesh
mdmosfikurrahman.cse@gmail.com

Musfikur Rahman Foysal
Department of Computer Science and
Engineering
Daffodil International University
Dhaka, Bangladesh
foysal15-8489@diu.edu.bd

Nazmun Nessa Moon
Department of Computer Science and
Engineering
Daffodil International University
Dhaka, Bangladesh
moon@daffodilvarsity.edu.bd

Fernaz Narin Nur
Department of Computer Science and
Engineering
Notre Dame University
Dhaka, Bangladesh
fernazcse@ndub.edu.bd

Abstract— The popularity of food delivery increasing day by day. This work focuses on how to food delivery of the easiest way in the train journey. In the modern-day, almost everyone trying this online food. Otherwise, in this pandemic, the demand for online food delivery is sky scraping. So, how it see when everyone is going to travel in long-distance by train, then if they can be trying healthy and delicious food from a restaurant? In train journeys, travelers cannot get good quality food or the food price is high. By using this android application, travelers can fulfill their food demand. This thesis focuses on the best way to supply food during the trip by rail. Almost all of us try this internet food today. In this research work, there have three types of modules like one application for traveler means customer, one application for riders' means deliveryman and another one is for restaurant authority. In this system customer can get good food, deliveryman gets a job and restaurant can increase their profit. The application needs to be responsive as the application can be accessed through any device with different sizes of screens. In this work, the firebase database and native android framework are used. Android studio application is used for writing code for the main body of this application. The food delivery apps have simple and easy features. So, it is easy to use for everyone. The new take-way system can make it easier and ensure that maintaining quality.

Keywords— *Android Application, Food Delivery, Design, Model, Customer, Restaurant.*

I. INTRODUCTION (HEADING 1)

The application sets to improve the online food ordering system for the train. The Online food ordering system is appearing to override the problem of the trouble of delivery and payments method. Nowadays people are reaching online businesses. Using an online service online service, it helps to save time and effort for both customers and salesman. In the Analyzing predictors of customer satisfaction, it shows that Regression analysis showed the customer satisfaction of several significant predictors, including Bank staff ($\beta = 0.18$, $p < 0.001$), bank guidance [$\beta = 0.004$, $p < 0.01$), service product variation ($\beta = 0.03$, $p < 0.05$), account access ($\beta = 0.11$, $p < 0.001$), availability of ATM cash ($\beta = 0.11$, $p < 0.001$) and banking staff knowledge ($\beta = 0.40$, $p < 0.001$) [12]. Particularly 18 adversaries and the structural characteristics of social media from previous work were highlighted in the examination of the 'Impact of social media on the learning behavior for sustainable education'[13]. The results of an Education Interventional ($p=0.011$) have been improved by

the ANCOVA. The robust ANCOVA indicated the effective intervention in individuals with low to intermediate knowledge levels, perception of sensitivity, severity of perception, obstacles felt and self-efficacy [14]. The results of the current study of internet addiction and its effects on mental health indicated that the mean total score for internet dependence and mental health students corresponded to 3.81 ± 0.88 and 2.56 ± 0.33 [15]. For using the application set no knowledge is need. Almost every organization are trying to overcome category, order, food item list, delivery, payment, etc. But every online food delivery system maintains different types of methods. When the Online food delivery system is starting for the first-time people are confused about their food quality. Many people or customers are worried about food quality. But when the online food ordering system is popular day by day the customers have become tension-free. Generally, people order food by reference of a friend or nearest relative. We have such thoughts after thinking about everything.

And the customer can order by seeing the restaurant menu book. If the customer places the order, then it shows in the delivery man application, then the delivery man going the restaurant to order the customer's food, and when the restaurant makes or packing this order the delivery man reaches a customer. Then the customer gives payment to the delivery man.

II. LITERATURE REVIEW

In a research work done by Leong et. al. [1] Food Ordering System Using Mobile Phone. This research work is mainly developing for computerized and mobilized system which can be used for ordering food system of food and beverage industry. Most food and beverage industries work manually. Here done all work by manpower, this system it needs a huge amount of paperwork which is not effective. For these reasons here occur human error due to manual work. It is also reducing time, minimizing human error, and provides good quality food. In a research work done by Md. Muminur et.al. [2] Implementation of Responsive Online Food Ordering Application with Social Media Integration. In this research work build an online application for a restaurant which helps customer to order in online food. Customers can be accessed through any kind of device with different sizes of screen. Here Facebook API is attached to the application. So, customers can attractive by seeing their friends, relatives review on

Facebook. For attaching Facebook API customers can log in to their Facebook account. It is a web-based online application. In a research work done by Michael Yosep et.al. [3] Mobile Food Ordering Application using Android OS Platform. This research work mainly focuses on New Order, Order History, Restaurant Profile, Order Status, Tracking Order, and Setting Profile features. This food ordering application is based on Android for customers and delivery man, and a website for restaurant authority and admin. It helps the customer for easy order, giving information details for consumer, it helps the restaurant to take the order, and help delivery man while doing delivery. In this research work done by Yang et.al. [4] Mobile food ordering Application. Here the application provides current food information like quality, price, categories, quantity, and customers can order food with two platforms. The application is three modules: Background Management Platform, Website Foreground Public Page and Android Application. This is an android based framework. The Background Management module in this android application was designed for the admin. The Admin will be able to handle food dishes, dish orders. In this research work done by Ashutosh et. al. [5] Digital Ordering System for Restaurant Using Android. In this research work, they focus on the Digital food ordering system. This system increases the quality and speed of service. Here an application of integration of hotel management systems by web services technology is presented. Digital Hotel Management integrates lots of systems of hotel industry such as Ordering System Kitchen Order Ticket (KOT), Billing System, Customer Relationship Management system (CRM). In this paper Cai et.al. [6] focused on the mindset matters in purchasing food from online delivery system during the pandemic. The project explores how interplaying contractual mindsets and message bodies affects consumption intentions in relation to online food supplies during a pandemic, applying theory of construction and regulatory focus theory. In this research Zhao et.al. [7] proposes a broad model of integration of UTAUT, ECM, and TTF with the confidence factor, as well as an examination of 532 valid FDA users' intention to continue to use FDAs during China's COVID-19 pandemic. During the COVID-19 pandemic the statistical results and discussion indicate that the satisfaction is the most significant factor, and that task technology is perceived to be fit, trustful, performance-expected, socially affected and confirmed. The purpose of the study of Al Amin et.al. [8] was to determine factors that influence consumer behavior and consumer continuing intention in the context of the COVID-19 pandemic to use mobile food delivery systems (MFDA). They investigated the impact of compliance with behavioral isolation and continued intention to use MFDA on food safety, food supply hygiene, subjective standards, catering, and behavioral control based on the theory of planned behavior. This study included theoretical situational factors which could guide practitioners to use MFDA at COVID-19. The aim of this study of Suhartanto et.al. [9] is to investigate conduct intentions towards the Islamic Bank, including three key factors: religiosity, trust, and image across customers and non-customers. This study offers Islamic bank managers the opportunity to increase customer and non-client behavior. Yeo et.al.[10] study aims to investigate the structural connection between the motivation of convenience, usefulness after use, hedonic motivation, price-saving orientation, a time-saving orientation, prior online buying experiences, consumer attitudes, and behavioral intentions for OFD services. It presents an integrated theoretical research model based on the

IT continuity contingency framework and extended model. The major objective of this research of Cheng et. al. [11] is to build quality OFD-service (OFD) quality, based on qualitative and quantitative research methods and incorporates important service elements retrieved from internet big data analytics (IBDA) (OFD-SERV).

III. METHODOLOGY

A. Business Process Modeling

In this figure, we can see the business process modeling for this application. Here a customer wants to order for a dishes. When customers check the menu and confirm an order it will send in a rider application. If the rider gets a notification for the customer confirming the order, then he accepts the order and sends the order to the restaurant. After finishing every process in a restaurant like taking orders, cooking order and packaging order then again rider take the food from the restaurant. Then rider provides the food to the customer and the customer makes his payment for the food and delivery charge. The business process model is shown in Fig.1.

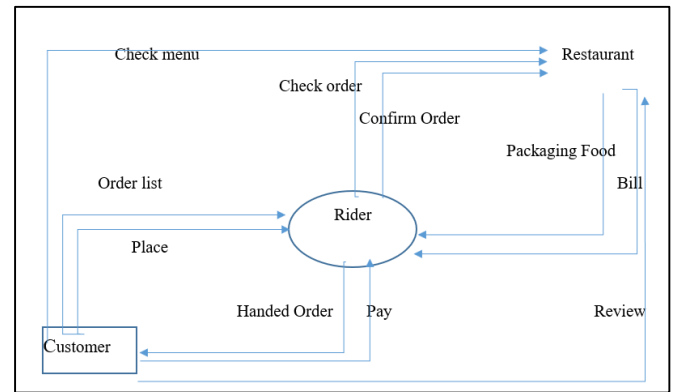


Fig.1. Business Process Model

B. Requirement Collection and Analysis

Requirement analysis is an important part of research work. We know there are two types of requirements as Functional requirement and non-functional requirement. Functional Requirement is a process of how the application performs and the non-functional requirement defines the behavior and efficiency of the application.

1) Functional Requirement

- The android application must have a signup and login option for the user.
- The application must have added menu and edit menu option.
- The application must have login and signup with a Gmail address and password.
- The android application must have a shopping cart for an online food ordering system.

2) Functional Requirement

- The android application must have a user interface.
- The user interface must have mobile-friendly.

3) Hardware and Software Requirement

a) Hardware

- Intel core i3
- 4GB ram
- 1TB hard disk drive

- Monitor
- USB port 3.0
- Wireless Connection.

In this research work, a computer with sufficient power is needed. For mobile application, database creation, and modification, the computer is requiring for the developer to have research work development such as coding. The Hardware requirement for the computer is shown in Table I.

TABLE I. HARDWARE REQUIREMENT FOR COMPUTER

Description	Minimum Requirement
Processor	1.1GHz or faster processor
RAM	512MB or more
Hard Disk	4GB of disk space available or more

Generally, we carry smartphones all time with us. For using the application in mobile phone some requirement is needed which is shown in Table II.

TABLE II. HARDWARE REQUIREMENT FOR MOBILE

Description	Minimum Requirements
Processor	500MHz or more
RAM	512MB or more
Storage	512MB or more
Internet Connection	Yes

b) Software

- Front-End: Android Studio Development kit.
- Operating System: Windows 8/ Windows 10/ Android 7.0 or more
- Programming Language: JAVA, XML.

For using the application some requirements are needed for software which is shown in Table III.

TABLE III. SOFTWARE REQUIREMENT

Description	Minimum Requirements
Mobile Operating System	Android Version 7 or more
Windows Operating System	Windows 7 or more
Interface Design	Firebase

C. Use Case Modeling and Description

In this use case model, we see that there were three actors in this application. Here the first actor is a customer, one is a railway rider and another is a restaurant authority.

1) For Railway Rider Module:

a) Login

Properties

- **Email**

The first condition of log into the application valid email address is needed. It is a unique address; it is not the same as one to another.

- **Password**

Also, without a valid password we cannot enter the main interface. Use case model is shown below Fig.2.

When Rider connects the background platform URL, a login interface will display. The rider uses the correct email address and password to log into the main interface. One rider can pre-set an email and password for when the application is initially started.

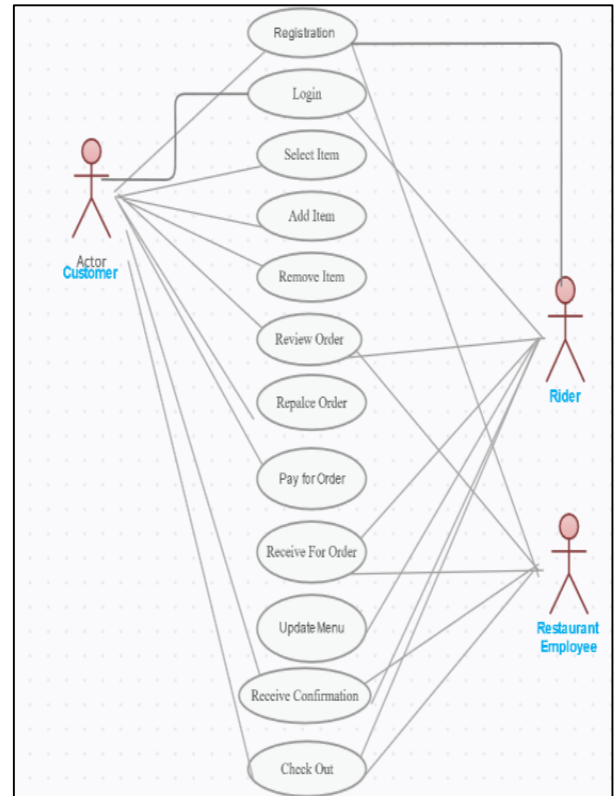


Fig.2. Use Case Modeling

b) Sign Up

Properties

- Name
- Surname
- Mail address
- Phone Number
- Password
- Confirm Password

When riders want to sign up for the application, first he needs to register by clicking the “Sign Up” button to enter the signup page. Then he filling in Name, surname, mail address, phone, password, and confirm password. Then the rider registers it successfully by clicking the “Confirm” button. If the rider registers it successfully, then a page will have displayed where he can inform the customer. If sign-up is not valid then a friendly message is notifying from the Sign-up page. Such, the name already exists or the email already exists. The study looks at the effect of CEO duality to accomplish firm performance by mediating the impacts of the innovative model of capital structure and competitiveness in the market. The study examined the proposed association between duality and CEOs' corporate success, using the GMM model, and shows the negative association between duality and CEO's business success. [21] This research on 'The Impact of Entrepreneurial Business Networks on Firms' Performance through a Mediating Role of Dynamic Capabilities' investigates how dynamic capabilities are related to the relation between the enterprise network and sustainable performance of small companies.

2) For Customer Module

a) Sign Up

When a customer wants to order dishes on the application, first he needs to sign up an account by clicking the “Sign Up” button to permeate the sign-up page, and then the customer filling in name, surname, mail address, phone number, password, confirm password. If the customer signs up successfully, a page will have displayed for the customer, if there is a fault in a register, such as a user email is already existing or a user name already exists then a friendly warning message will give from the sign-up page. In this sign-up function there is every information filed valid, then the sign-up successful.

b) Log In

After Sign up, the customer can log in with the correct Email address and password to enter the application to create an order.

c) View Dishes

To click in Restaurant, name the customer view the current dishes according to categories. Every dish item has two buttons: Order and Review. The Customer tap into the Order button and here he can add quantity means how many dishes he needs, the here customer can see the item price. After confirming the order the customer will have added to the shopping cart.

d) Shopping Cart

The dish items buy by the customer can be added to the shopping cart. Then the actor can view the item which his buy, price, quantity, and total price. Customers can update items and delete an item.

3) For Restaurent Module

a) Sign Up

When a customer wants to order dishes on the application, first he needs to sign up an account by clicking the “Sign Up” button to permeate the sign-up page, and then the customer filling in his name, surname, mail address, phone number, password, confirm password. If the customer signs up successfully, a page will have displayed for the customer, if there is a fault in the register, such as user email is already existing or user name already exists then a friendly warning message will give from the sign-up page. In this sign-up function there is every information filed valid, then the sign-up successful.

b) Log In

After Sign up, the customer can log in with the correct Email address and password to enter the application to create an order.

c) Add Food List

When a restaurant owner Click on Add food list and then will show a page where restaurant authority can add food dishes name, price, and quantity. After press confirm button the item name is listed in this application and the customer will see the food name.

4) Logical Data Model

The logical data model is one kind of relational table. That is a table with different entities. Here we have some relational tables like Customer, restaurant, rider, order, menu, food item. The table has some attributes like customer id, name, number,

password, food name, food id, menu name, menu id, etc. The whole logical model is shown in Fig.3.

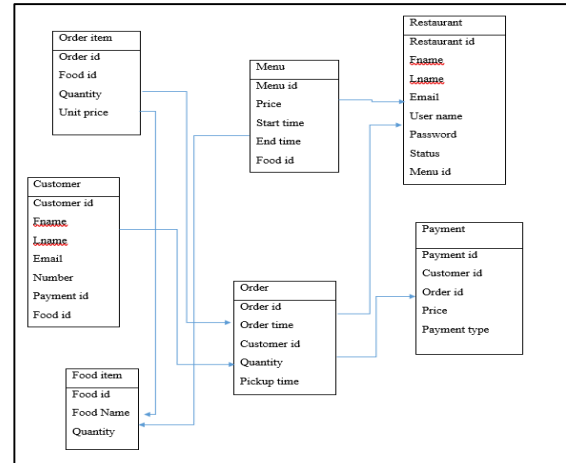


Fig.3. Logical Data Model

5) Design Requirements

Design Requirements is the functional characteristic that qualifies to convert the idea into a design feature. In this part, Author will describe the design requirement for of this application which is software requirements and hardware requirements. For implementation main body of the application software requirements are most important. Also, hardware is important for design and application. The software requirement list is given below:

a) Software

- Front-End: Android Studio.
- Operating System: Windows 8/ Windows 10/ Android 7.0 or more
- Programing Language: JAVA, XML.

The second most important part is hardware requirements. The hardware requirements list is given below:

- Intel core i3
- 4GB ram
- 1TB hard disk drive
- Monitor
- Internet Connectivity
- Android Smart Phone.

6) Front End Design

In the previous discussion, we have discussed a list of the requirement of the application. Now, in this section, we will discuss the all requirements for this research work.

a) Software Requirement

- Android Studio

Whereas the research work is an android based application so we choose android studio for building the main structure code. By using android studio, we make an android application for collecting data from the user

- Operating System

An operating system is the most important software for run a computer. Computer memory manages done

by the operating system. An operating system manages the central processing unit for a computer.

- **Programming Language**

A programming language is used to automate, assemble, and maintain data and information. It is most useful and essential for building software. By programming language, an application or system will run and work.

b) Hardware Requirement

- **Monitor**

When users sign up for the application those data will store in the firebase database. For writing and run the code of research work, we need a computer. So, a monitor is needed for the research work.

- **Internet Connectivity**

When the user signs up and logs in the application internet connection must need. Because of the information of the user store in firebase. Without an internet, connection customers cannot order food from this application.

- **Android Smart Phone**

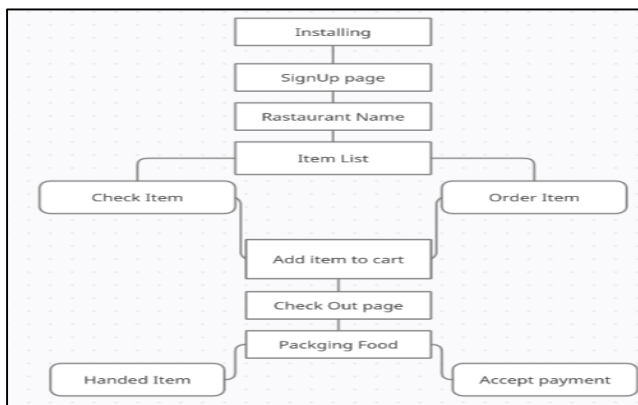
For installing the application, an android smartphone needs. After installing, the user can see a sign-up page. After completing, the sign-up users can use the application easily.

7) Back End Design

In this discussion, we are going to discuss the back-end development for the research work. We will discuss the flowchart diagram of the research work. The back-end design is shown in Fig.4.

Fig.4. Back End Design for this Application

8) Interaction Design and UX



Interaction Design is a technique using that developers are model their application and also system interface more user-friendly. Interaction design has many types like Visual portrayal, Words, time, and conduct. I'm trying my best to build the application with the above interaction. Visual presentation is an important part of any application. The author can try to build the application more user-friendly. Author use here firebase for database and android native framework which can make interface smooth. The author design the application interface is very simple for the user. My application has three modules. Every module has Sign Up and login interface. By using the sign-up button every module user must save their information. After register in

sign up the user can enter inside their module by using the login option. Here also the Logout button. By improving the application user can get a better response from the application. UX or User Experience is also a more essential part of any application. The author was checking a survey in online and real-life and get positive feedback from them whose are using these types of application which are made before. Then Author runs tests by some people. The application is faster and comfortable.

9) Test Result and Report

In this section, we will discuss the test result and report for this application.

a) Security

For an android application, security is one of the most important parts. As it is an android based online food ordering system here saves different types of personal information. Again there needs a location system for ordering food. For ensuring the security of the application different methods are used. Laravel has been used for hashing functions for passwords. Laravel handles all password encrypting and decrypting automatically.

b) Responsive UI

Any kind of device screen is responsive to a user interface to the application. Android native Framework has been used to develop the user interface system. Mobile, tablet laptop, or desktop any kind of screen enables the UI of the android application.

IV. IMPLEMENTATION

A. Implementation Requirement

In a previous discussion, the Author explained this research work in an android based application. The author needs to use computer language, framework, database, and XML for the design interface. Author use here JAVA language to build the application architecture. Author compose code by Android Studio. My full research work is work by following this code. Here, the author uses the android native framework for designing. Basically, for an android application using the firebase database. So, the Author also uses firebase for smooth connectivity. For using this application must need an internet connection. Although customers and delivery man can contact their phone customer create order by the application so that internet connection is a must. Hardware and software equipment combination and good arrangement will make the application smarter.

B. Database Implementation

The application will need a database system to store a huge amount of data for support in order system. Here, the Firebase database is chosen is for the proposed database because of firebase database is well known for the android database management system. Otherwise, the firebase database system provides an easy understanding graphical interface for a software developer with a client program. Developer can interact with the firebase database system user-friendliness with the client program. The above Author explained about the database system. We also explained the connection between one module to another module.

C. Implementation of Front End Design

Here in this part Author explain Briefly Front end design of this application. The user interfaces were created with XML

code. I'm using an android application for the demand of the user. Because now a day most people use an android smartphone. For making my android application Author uses android studio and using JAVA language for a programming language to build the application. In my application has three modules, one for the customer, one for the rider, and another one for the restaurant. Each module has a signup page. On the signUp page, the user must add their information such as Email address, Phone number, address, Name, Surname, Password, and confirm password. After completing the sign-up successfully the user can log in with an email address and password. For the customer module customer can see the restaurant name and then check the dish name, add dishes quantity, customer can review their food in this application. When a customer adds an order, it going on the Railway Rider application. The rider checks the order and then he confirms the restaurant for making this order. After processing the step the restaurant authority packaging the food for the customer and handed in Rider. After taking the rider this food, he/she provides the food to the customer. When the customer gets the ordered food then he/she make the payment of food and delivery to Rider.

D. Implementation of Interaction

The interaction design of a system also one of the most important ingredients. Interaction design is associated with apps or web site. It applies to anything that the user can touch. Generally, interaction means communication between two or more things. This communication can be a human to human or human to the application or any system. For the customer module, the Customer must need to fulfill the sign-up and login for registration of this application. Then Customer can view the restaurant name and check the menu. Then the customer can add his/her favorite item to the cart. For the Rider module, Rider also fulfills the sign-up and login page for registration. The rider can set here is distance or root. Then if the customer sends, an order request rider will confirm it. Then the rider confirms the order in the restaurant.

E. Implementation of Interaction

After the application is developed, it needs system testing. The required application needs to install the inappropriate device for the testing phase.

1) Unit Testing

Unit testing is the first phase of testing any application or system. The developer can identify errors and bugs easily by doing unit testing. The system process and expected result need to testing valid and invalid input. The unit testing for the login system is shown in Table IV.

TABLE IV. UNIT TESTING FOR LOGIN SYSTEM

Event	Attribute And Value	Expected Result	Result
Verify that ID and password that enter by the user and match the data in the database. Verify the invalid ID and password that enter by the user and match with the data that store in the database. Verify the situation that user does not enter any value	Login ID: valid ID Password: valid Pass	Login successfully. Login failed and prompt out the error message to the user.	Pass Pass
	Login ID: valid ID Password: invalid Password OR Login ID: invalid ID Password: valid Password	Login failed and prompt out the error	Pass
	Login ID: null Password: null		

2) Functional Testing

After complete unit testing, Functional testing will begin for testing the developed application. The system application function will be testing in functional testing. Functional testing is performed with the whole perform task with expected results. The functional testing is shown in Table 5.2.

TABLE V. UNIT TESTING FOR LOGIN SYSTEM

Event	Attribute and Value	Expected Result	Result
Login as "Customer"	Log in with customer information	Successfully Login.	Pass
Login as "Rider"	Login with Rider information.	Successfully Login.	Pass
Login as "Restaurant"	Login with Restaurant information.	Successfully Login.	

3) System Testing

System testing is the critical testing procedure. System testing must be condiment by the system developer before the application is released. System testing of software and hardware testing is condiment on a system that is complete. System testing is not only testing the design of the application but also tests the behavior of the application. It has expected results to solve errors and bugs.

V. DISCUSSION & CONCLUSION

The android application "Food Delivery Application for Train" has been effectively planned. This Research work contains three modules, Railway Catering, Railway Rider, Railway Restaurant. The three platforms were implemented by firebase for database, XML for creating user interface design. The native android framework has been used in the android application. This research work was a typical combination of every module of the application. By this research work restaurant owner, rider, customer everyone can be benefitted. The main function of this research work has been implemented. It helps to improve the efficiency of restaurant management. Customers can view the page of item name; customers can add quantity for their favorite dishes. It reduces the time consumed for each transaction. After All, it will be easier and comfortable for train travelers.

VI. PERCEIVED CHALLENGES

In this modern era technology e.g. IoT [12-16], Cybersecurity[17], Machine Learning [18-26] and so on plays a vital role where in this research work, the main attenuation is the internet connection. Without an internet connection, it will not operate. Because if a customer wants to order must need mobile data or a Wi-Fi connection to send data in the restaurant database. Next, the application can be developing for IOS and blackberry users. Therefore, the user of blackberry and IOS operating system, they can use the application take orders easily. And also the segmentation of users will be wider. In the next development, this android application can be developing for payment methods. Added mobile banking, Visa card, Master card for easily purchase restaurant charge by customer. In the next development, the android application will implement with social media like Facebook, Twitter, Instagram, Linked In ETC for getting restaurant and food information

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